1 **(a)** Plotting the remaining 6 points correctly. **[P2]**

(Allow P1 for 4 or 5 points correct).

(b) Negative **[B1]**

(c) A ruled straight line of best fit drawn by eye, with a roughly equal number of points on each side. **[B1]**

[Total Marks: 4]

2 **(a)** All 8 points plotted correctly **[B2]**

(5, 6, or 7 points plotted correctly [B1])

(b) Positive **[B1]**

(c) Ruled line passing through the mean point $(45, 45)$ and fitting the trend of the data **[B1]**

(d) Value read correctly from their line of best fit (typically $42 - 48$) **[B1]**

[Total Marks: 5]

3 **(a)** $y = 5.96x + 34.7$ (or $m = 5.96$, $c = 34.7$) **[B2]**

(Allow B1 for either $m = 5.96$ or $c = 34.7$ correct to 3 sf)

(b) Substitution of $x = 7$ into their equation from (a): $5.96(7) + 34.7$ **[M1]**

76.4 (or 76) **[A1ft]**

(c) The value $x = 15$ is outside the range of the given data (extrapolation). **[B1]**

[Total Marks: 5]

4 **(a)** Points $(7, 5)$ and $(9, 1)$ correctly plotted **[B1]**

(b) Mean age = 4.5 **[B1]**

Mean price = 11 **[B1]**

(c) Mean point $(4.5, 11)$ plotted accurately **[B1]**

Ruled line of best fit passing through $(4.5, 11)$ **[B1]**

(d) Negative (or strong negative) **[B1]**

(e) Value read correctly from their line of best fit (typically $2 - 4$) **[B1]**

[Total Marks: 7]

5 **(a)** $\bar{x} = \dfrac{2100}{12} = 175$ **[B1]**

$\bar{y} = \dfrac{924}{12} = 77$ **[B1]**

(b) Substitution of $(\bar{x}, \bar{y})$ into the equation: $77 = 0.85(175) - c$ **[M1]**

$c = 148.75 - 77 = 71.75$ **[A1]**

(c) Predicted mass $= 0.85(180) - 71.75 = 81.25$ **[M1]**

New mean $= \dfrac{924 + 81.25}{13} = 77.3$ (or 77.32...) **[A1]**

[Total Marks: 6]

6 (a) Mean temperature = 21.25 **[B1]**

Mean ice creams sold = 80.25 **[B1]**

(b) $m = 4.28$ **[B1]**

$c = -10.6$ **[B1]**

(c) $y = 4.28 \times 24 - 10.6$ **[M1]**

92 (or 92.12) **[A1]**

(d) Unreliable because 42°C is outside the range of the given data (extrapolation) **[B1]**

[Total Marks: 7]

7 (a) Positive **[B1]**

(b) Single straight ruled line passing within the bounds of the data points, with roughly equal numbers of points on either side. **[B1]**

(c) 64 (Accept values between 62 and 67 strictly based on the student's line of best fit) **[B1]**

[Total Marks: 3]

8 (a) All three points $(6, 81)$, $(7, 88)$, and $(8, 100)$ plotted correctly **[B2]**

(Allow B1 if only 1 or 2 points are plotted correctly)

(b) $\bar{x} = \dfrac{35}{7} = 5$ **[B1]**

$\bar{y} = \dfrac{490}{7} = 70$ **[B1]**

(c) Mean point $(5, 70)$ correctly plotted (allow a distinct mark e.g. a circle around the cross) **[B1]**

Single straight ruled line passing exactly through their mean point, following the trend of the data **[B1]**

(d) 75 (Accept values between 73 and 77 strictly based on the student's line of best fit) **[B1]**

[Total Marks: 7]

9 (a) (i) $m = -5.82$ (accept -5.81 to -5.82) **[B1]**

$c = 27.0$ (accept 27.0 to 27.1) **[B1]**

(ii) The decrease in fuel consumption (in km/l) for every 1 litre increase in engine size. **[B1]**

(b) Substitution of $x = 1.6$ into their equation: $-5.816(1.6) + 27.0$ **[M1]**

17.7 (accept 17.6 to 17.8 depending on rounding of m and c) **[A1]**

(c) Because 5.0 is outside the range of the given data (extrapolation). **[B1]**

[Total Marks: 6]